MATH 350-2 Advanced Calculus

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Outline

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1. Real Analysis Lecture 12
   - Limits of sequences
   - Cauchy sequences

Definition of a limit

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Back in Calc. I:

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$$\lim_{n\to\infty} x_n = L \quad \text{means}$$

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Metric space version:

$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } d(x_n, L) < \epsilon \; \forall n \geq N.$$

We say $\{x_n\}$ **converges** to L and L is the **limit of the sequence.**

Example

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Then for all $\epsilon > 0$, there must exists an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $N \geq n$.

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Then for all $\epsilon > 0$, there must exists an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $N \geq n$.
Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $N \geq n$.

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Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $N \geq n$.
With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.

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Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $N \geq n$.
With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.
This means $x_n = 1/n = L$ for all $n \geq N$, which is a contradiction! $\square$

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Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n > N_1$.

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Likewise, there exists N_2 such that $d(x_n, M) < \epsilon$ for $n > N_2$.

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This is a contradiction. $\square$

Existence of limits

Theorem

If a sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$, then the range

$$X = \{x_1, x_2, \ldots\} \subseteq M$$

is a bounded set and L is an adherent point of X.

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When the range of a sequence is bounded, we call the sequence $\{x_n\}$ bounded.

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If we take

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then $d(x_n, L) \leq R$ for all N.
Therefore $X \subseteq B_M(L, R)$ and in particular X is bounded. $\square$

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Theorem

If S is a subset of a metric space (M, d) and $L \in M$ is an adherent point of S, then there exists a sequence $\{x_n\}$ of elements of S which converges to L.

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For each $n \in \mathbb{Z}_+$, choose $x_n \in S \cap B_M(x, \frac{1}{n})$.

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We claim $\lim x_n = L$.

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To see this, let $\epsilon > 0$.

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It follows that for all $n \geq N$

$$d(x_n, L) < 1/n < 1/N < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves convergence. $\square$

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A sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$ if and only if every subsequence $\{x_{k(n)}\}$ also converges to L.

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Let $\{x_{k(n)}\}$ be a subsequence of $\{x_n\}$.

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To show it limits to L, let $\epsilon > 0$.

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Let $\{x_{k(n)}\}$ be a subsequence of $\{x_n\}$.
To show it limits to L, let $\epsilon > 0$.
Then there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d(x_n, L) < \epsilon$.
Moreover, $k(n) \geq n$ so $k(n) \geq N$ and therefore $d(x_{k(n)}, L) < \epsilon$.

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Moreover, $k(n) \geq n$ so $k(n) \geq N$ and therefore $d(x_{k(n)}, L) < \epsilon$.
Since $\epsilon > 0$ was arbitrary, this proves convergence. $\square$

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It is called **strictly decreasing** if $x_n > x_{n+1}$ for all n.
In any of these cases, we call the sequence **monotone**.

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A monotone increasing sequence $\{x_n\}$ of real numbers which is bounded above converges to $L = \sup\{x_1, x_2, \ldots\}$.

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Theorem (Monotone convergence theorem)

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A monotone increasing sequence $\{x_n\}$ of real numbers which is bounded above converges to $L = \sup\{x_1, x_2, \ldots\}$.
A monotone decreasing sequence $\{x_n\}$ which is bounded below converges to $L = \inf\{x_1, x_2, \ldots\}$.

Monotone sequences

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Suppose that $\{x_n\}$ is a monotone increasing sequence which is bounded above and let $X = \{x_1, x_2, \dots\}$.

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Since $\epsilon > 0$ was arbitrary, this proves x_n converges to L. $\square$

Outline

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then $d(x_n, x_N) \leq 2024$ for all $n \geq 1$.
Hence $X = \{x_1, x_2, \ldots\} \subseteq B_M(x_N, R)$, and is bounded.